SAKARYA ÜNİVERSİTESİ
Bilgisayar ve Bilişim Bilimleri Fakültesi
Bilgisayar Mühendisliği Bölümü

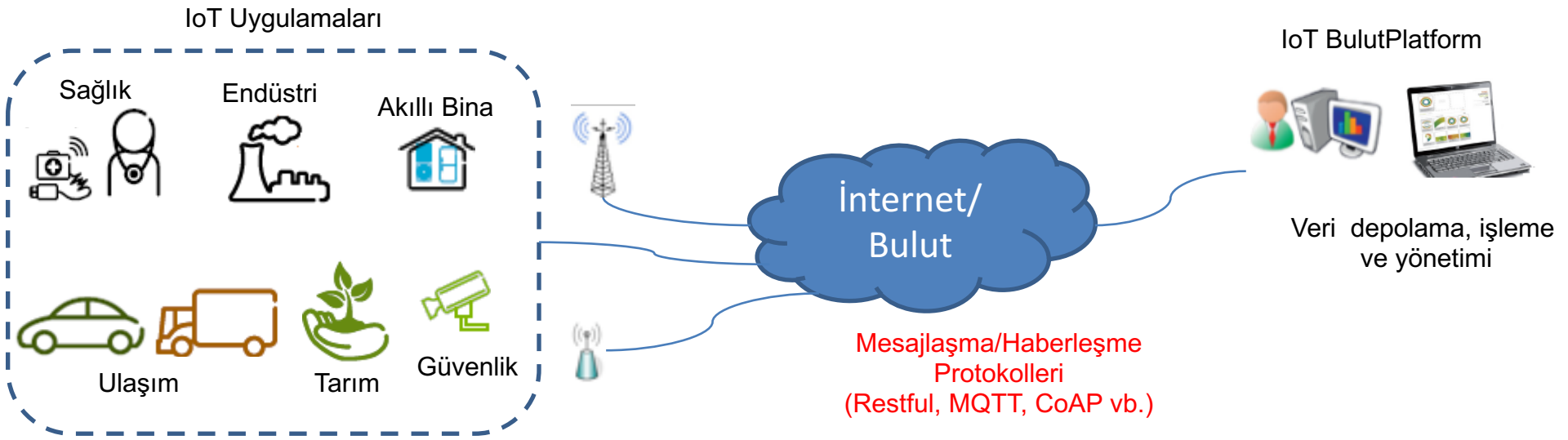
BSM 451
NESNELERİN İNTERNETİ VE UYGULAMALARI

(Internet of Things (IoT) and Applications)

NESNELERİN İNTERNETİ UYGULAMALARINDA KULLANILAN TEKNOLOJİLER

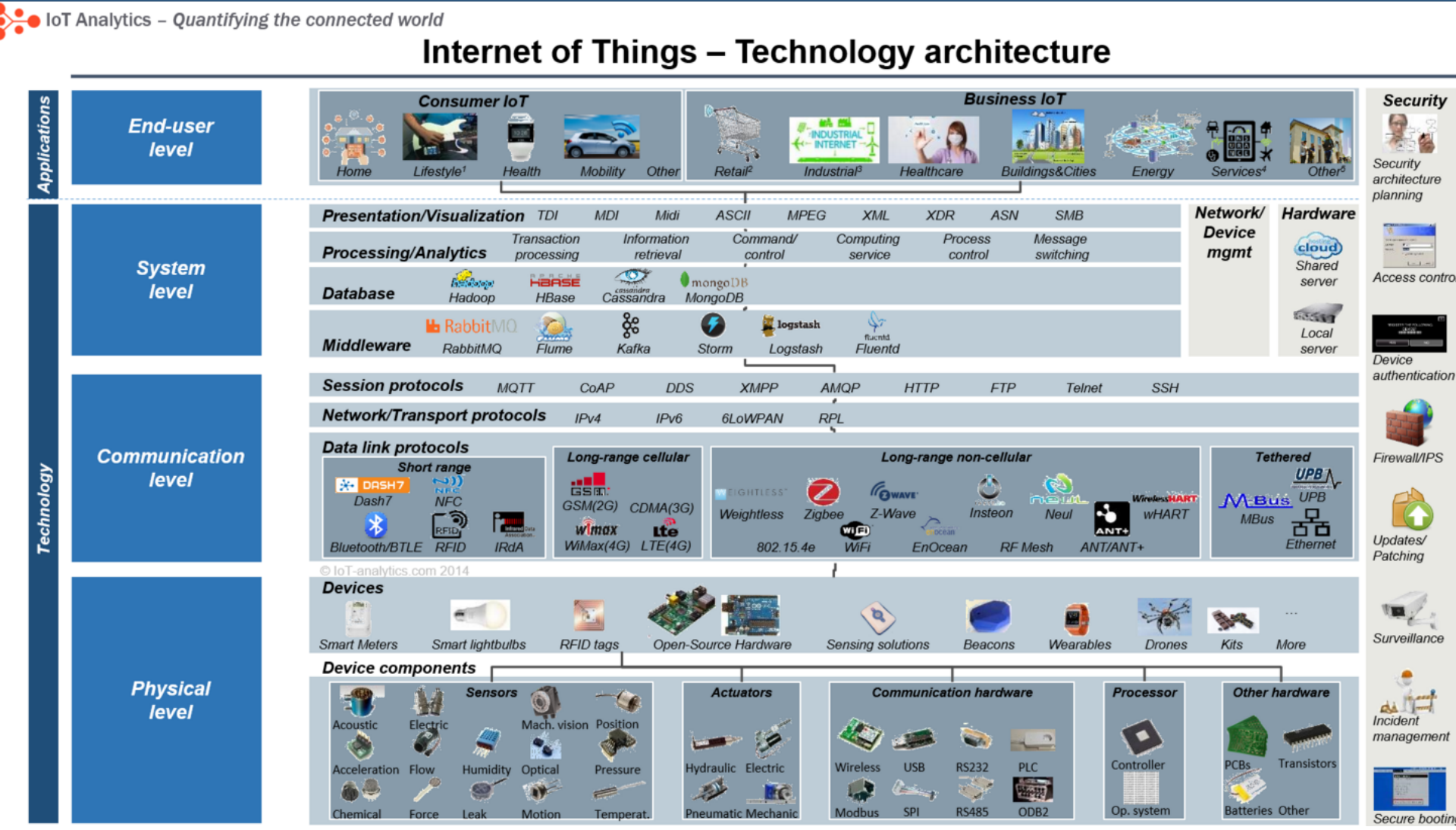
Doç. Dr. Cüneyt BAYILMIŞ

IoT Uygulamalarında Kullanılan Teknolojiler



IoT Yardımcı Teknolojiler
- RFID
- NFC
- Bluetooth Low Energy
- GSM
- GPS vb.

Nesnelerin İnterneti Teknoloji / Protokol Mimarisi



IoT Ağ ve Bileşenleri

- Nesnelerin interneti ağ bileşenleri

Bilgi Teknolojileri
(Information Technology, IT)

+

İşlemsel Teknoloji
(Operational Technology, OT)

+

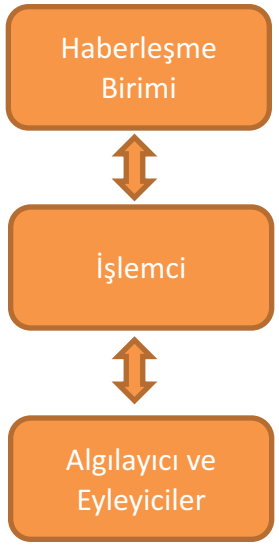
Akıllı Nesneler
(Smart Objects)

- İşlemsel teknoloji, doğru zamanda doğru kişi için doğru bilgi ulaştırma.

- IoT Akıllı Nesne



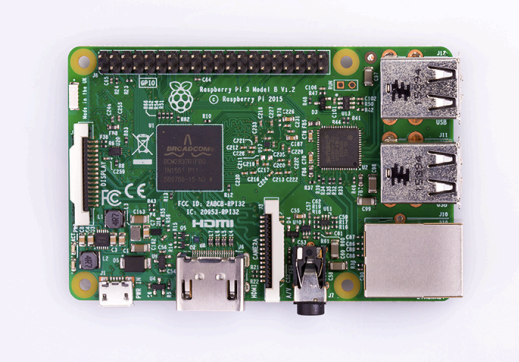
- Veri işleyebilen, akıllı karar verme yeteneğine sahip, birbirleri ve internet ile haberleşebilen fiziksel elemanlar.

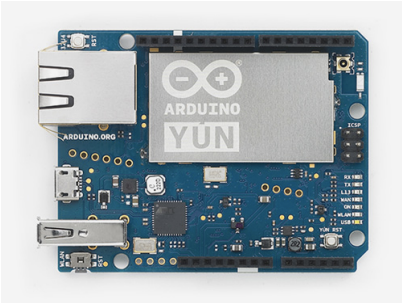


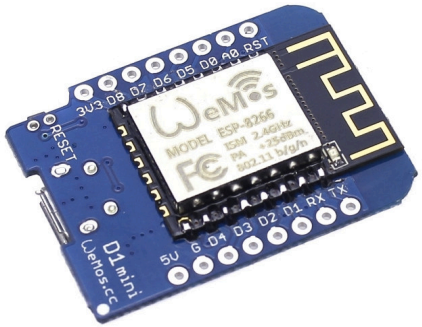


IoT Donanım Örnekleri

- Açık kaynaklı donanımlar

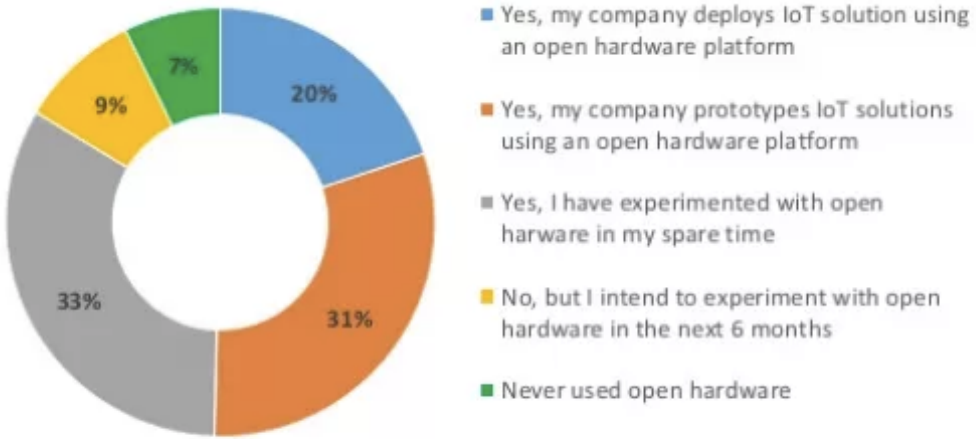






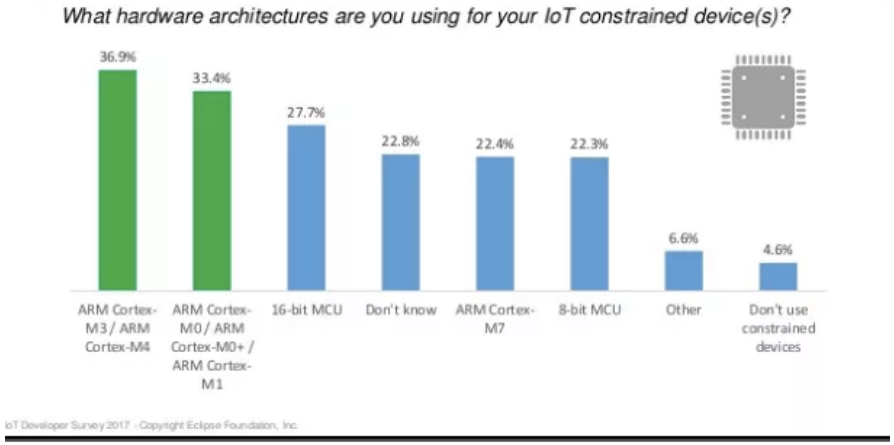
USAGE OF OPEN HARDWARE

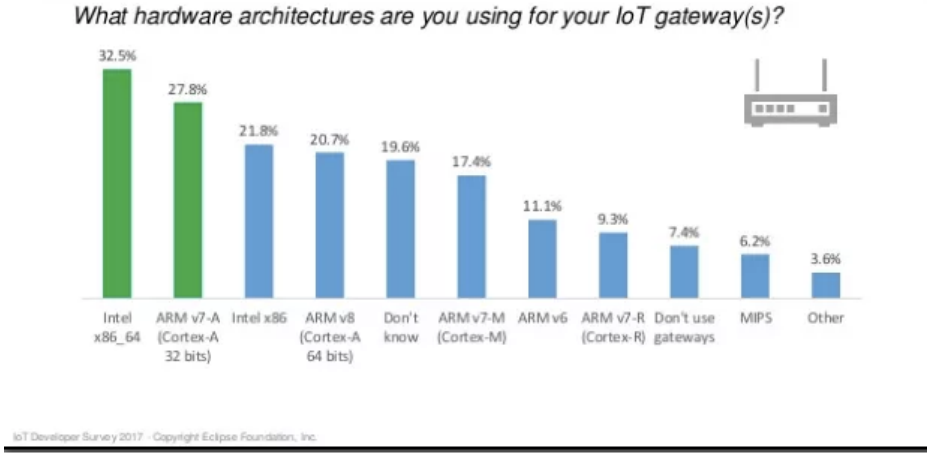
Have you ever used any open hardware platforms like Raspberry Pi, Arduino, BeagleBone, etc.?



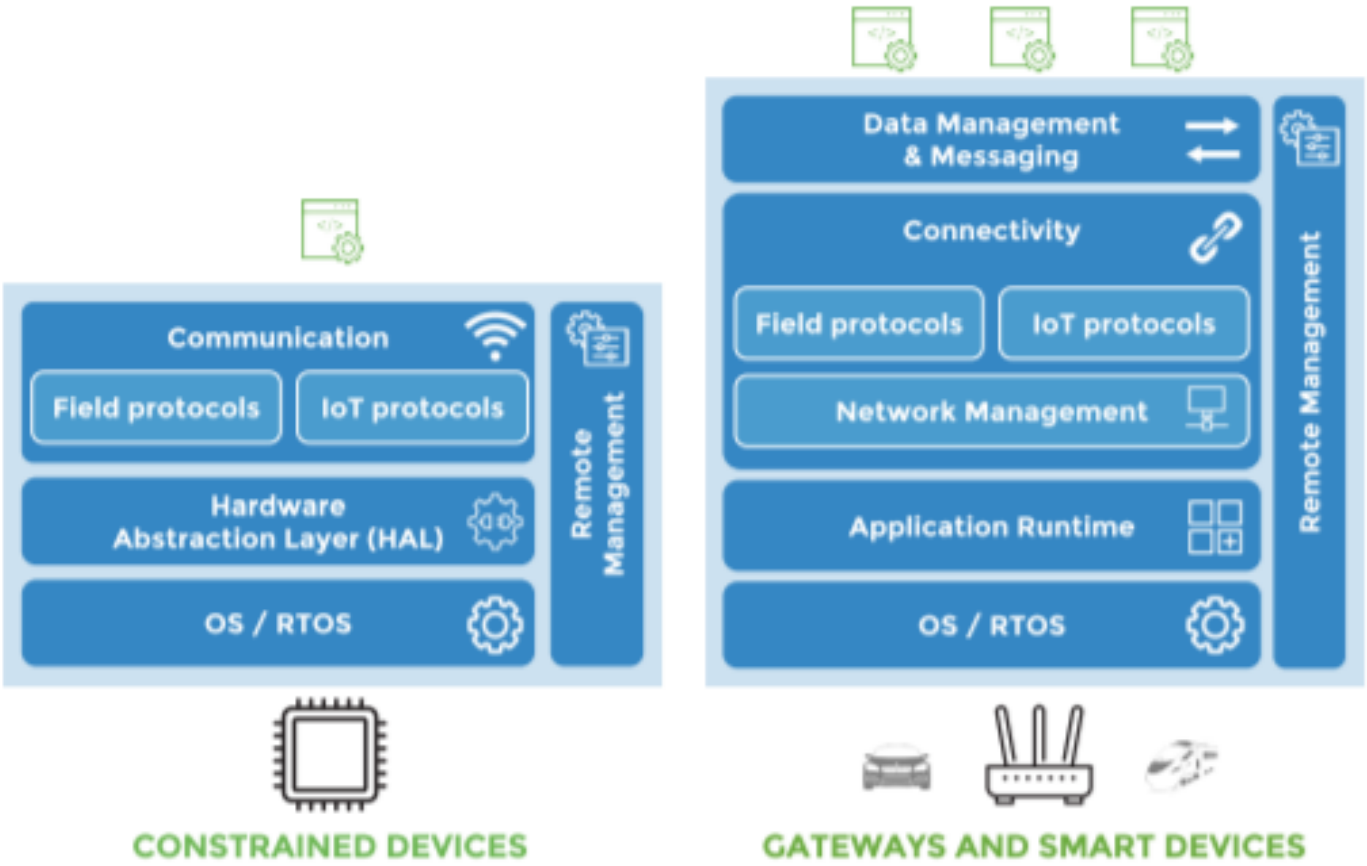
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IoT Donanım Mimarileri





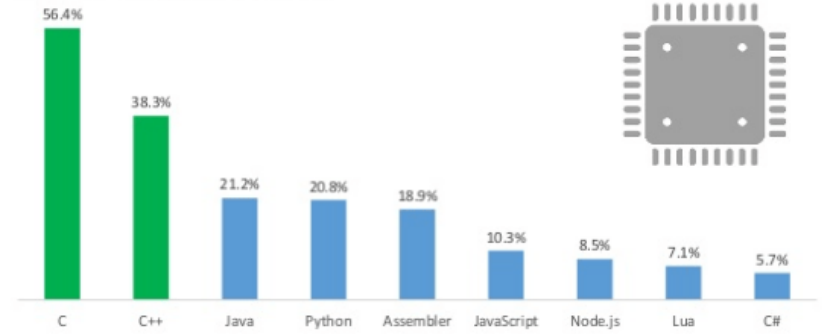
IoT Nesne Gömülü Yazılım Modelleri



IoT Nesnelerini Programlama Dilleri

Programming Languages – Constrained Devices

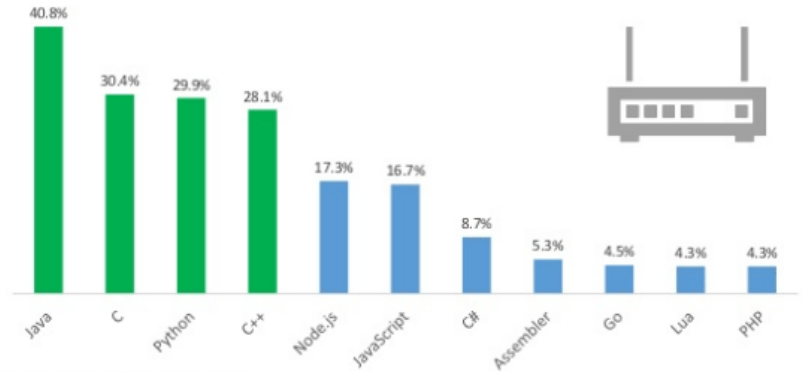
Which of the following programming languages, if any, do you use to build IoT solutions? (Constrained Devices)



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Programming Languages – IoT Gateways

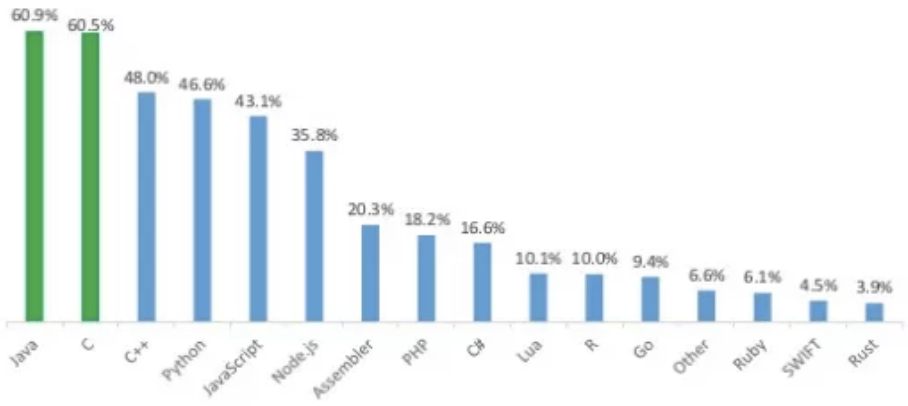
Which of the following programming languages, if any, do you use to build IoT solutions? (Gateways)



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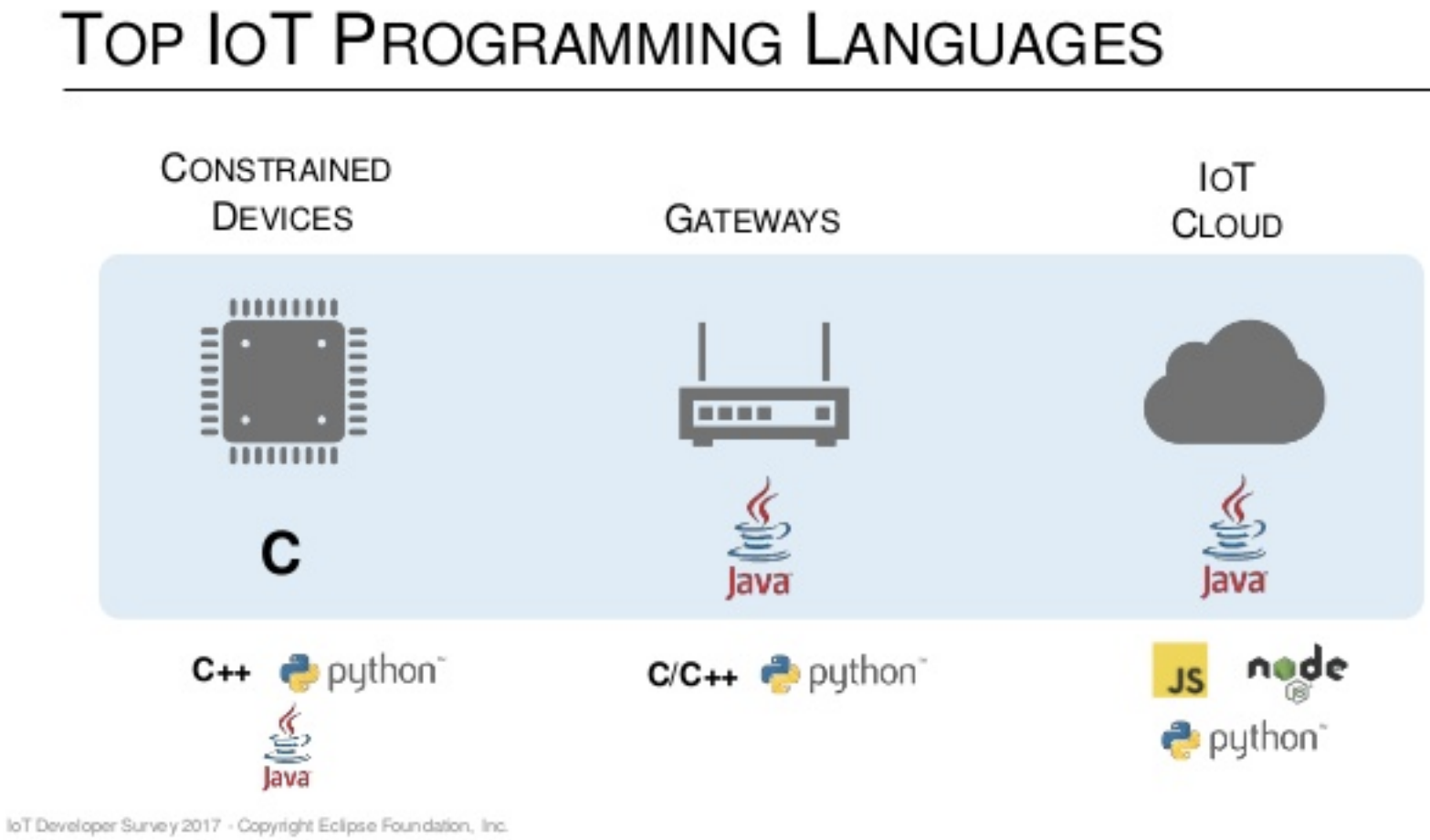
Overall Summary of Language Usage

Which of the following programming languages, if any, do you use to build IoT solutions?

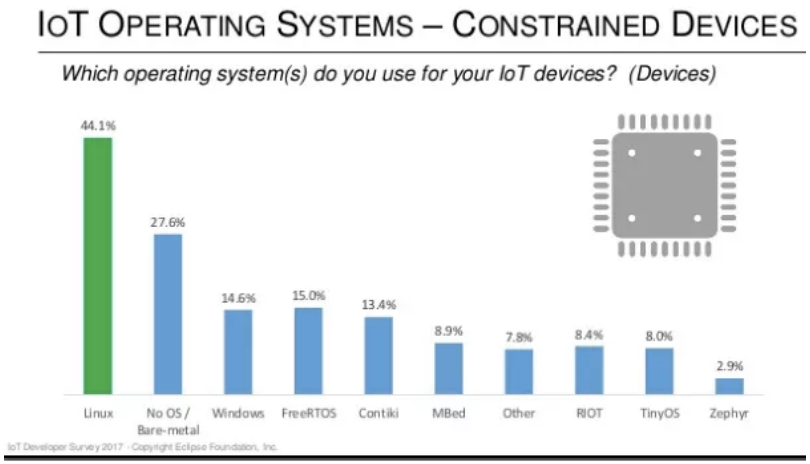


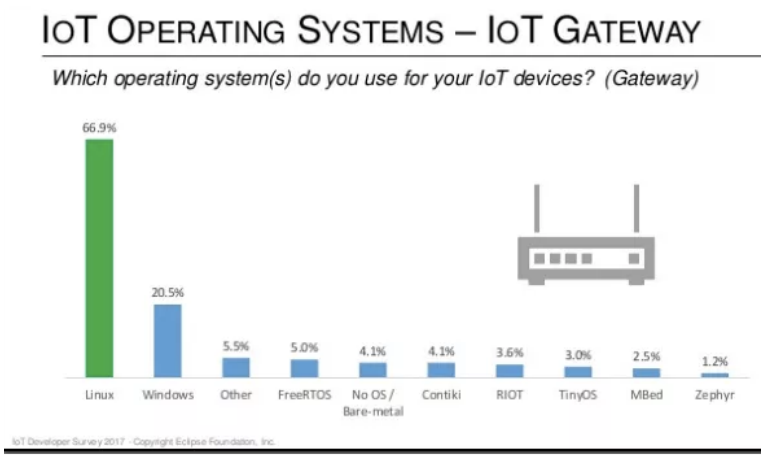
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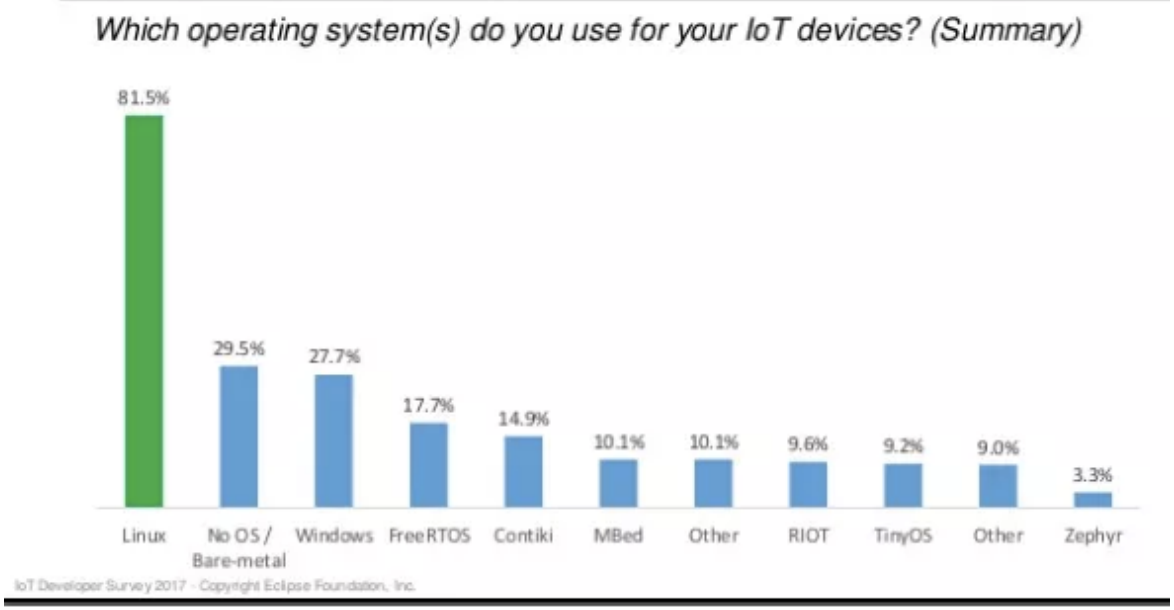
IoT Nesnelerini Programlama Dilleri



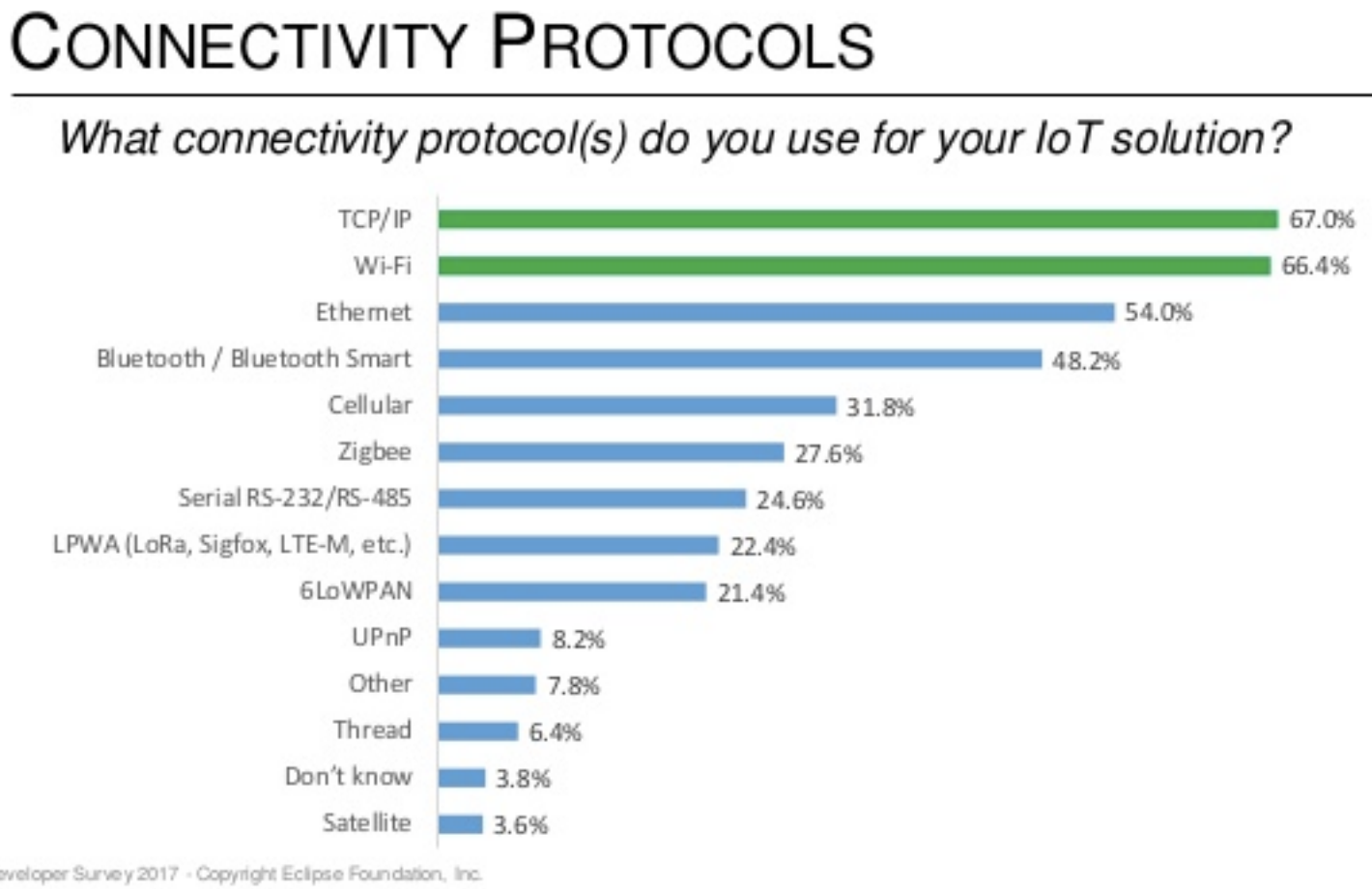
IoT Donanımlarında Kullanılan İşletim Sistemleri







IoT Bağlantı Protokolleri



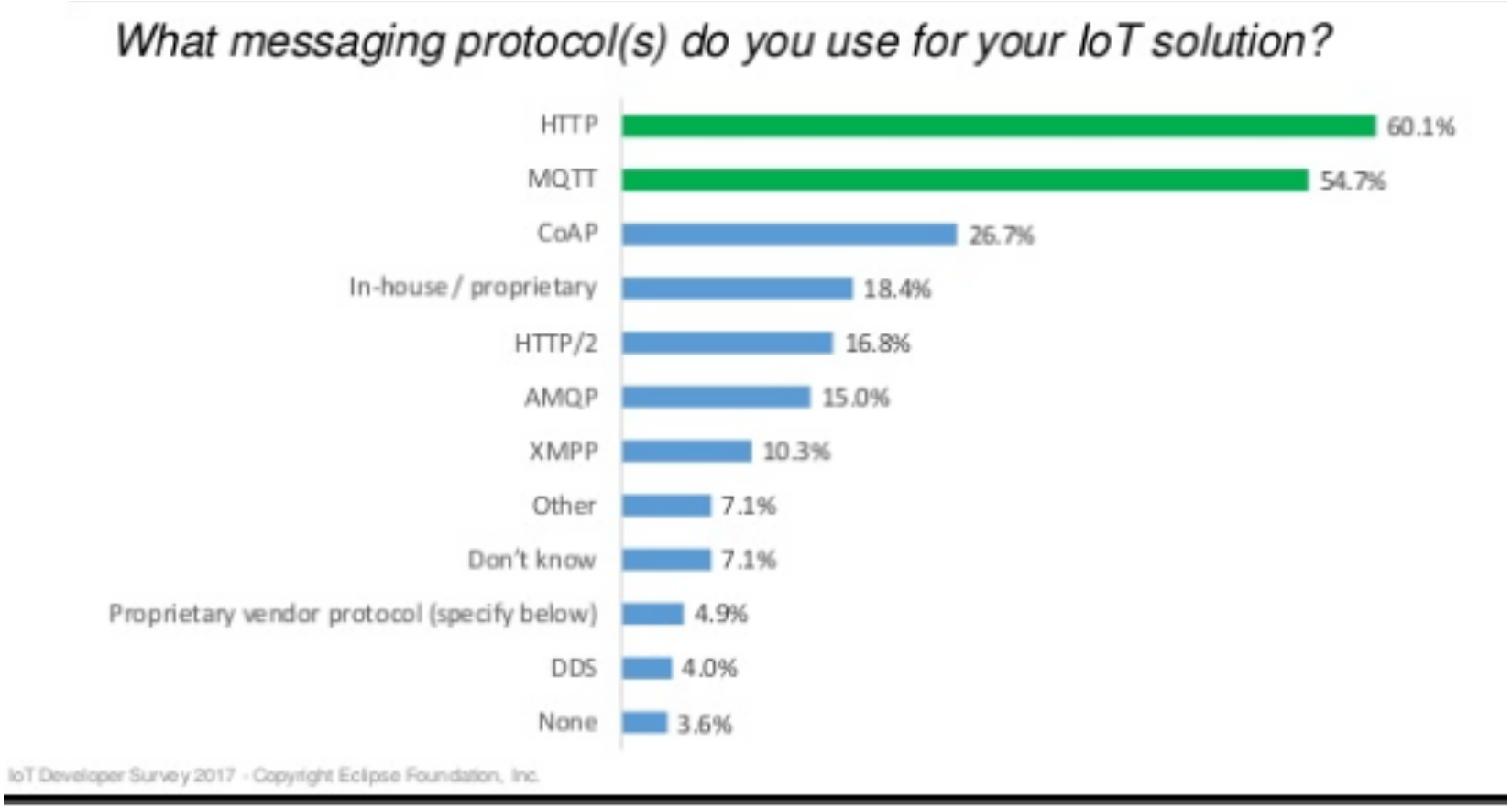
IoT Mesajlaşma/Haberleşme Protokollerine Genel Bakış

- Ağ üzerinden birlikte çalışabilen makineden makine etkileşimi/haberleşmeyi desteklemek için tasarlanmış yazılım sistemleri 'Web Servisleri' olarak adlandırılmaktadır.

- Web Servisleri kavramı, IoT konusunda uygulama/haberleşme protokolleri içerisinde ele alınmaktadır.

- Uygulama/Haberleşme Protokolü seçiminde kısıtlı donanıma sahip IoT cihazlarının kaynak kullanımı CPU, bellek vb. kaynaklarını efektif kullanabilmesi kriteri gözönünde tutulmalıdır. Bu nedenle IoT uygulamalarının hafif (lightweight) protokoller ile haberleşme ihtiyacı vardır.

Nesnelerin İnterneti Mesajlaşma Protokolleri



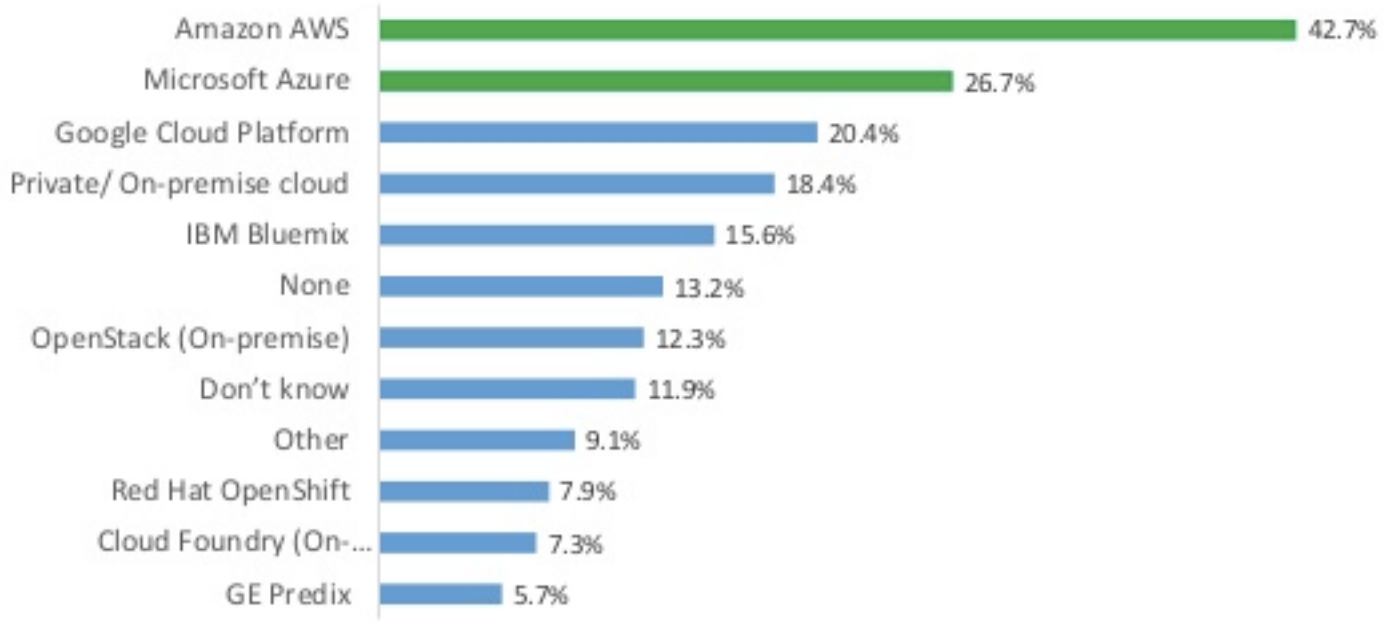
IoT Bulut Platformları

- IoT uygulamalarının doğası gereği nesneler tarafından üretilen/elde edilen verilerin internet ortamında depolanacağı ve görselleştirileceği/analiz edileceği ortamlara ihtiyaç vardır.
  - ThingSpeak,
  - adafruit,
  - Firebase,
  - TeMBoo,
  - IBM Watson IoT,
  - Microsoft Azure IoT,
  - Amazon Web Services (AWS) IoT,
  - ThingWorx IoT Platform,
  - Carriots,

IoT Uygulamaları İçin Bulut Servisler

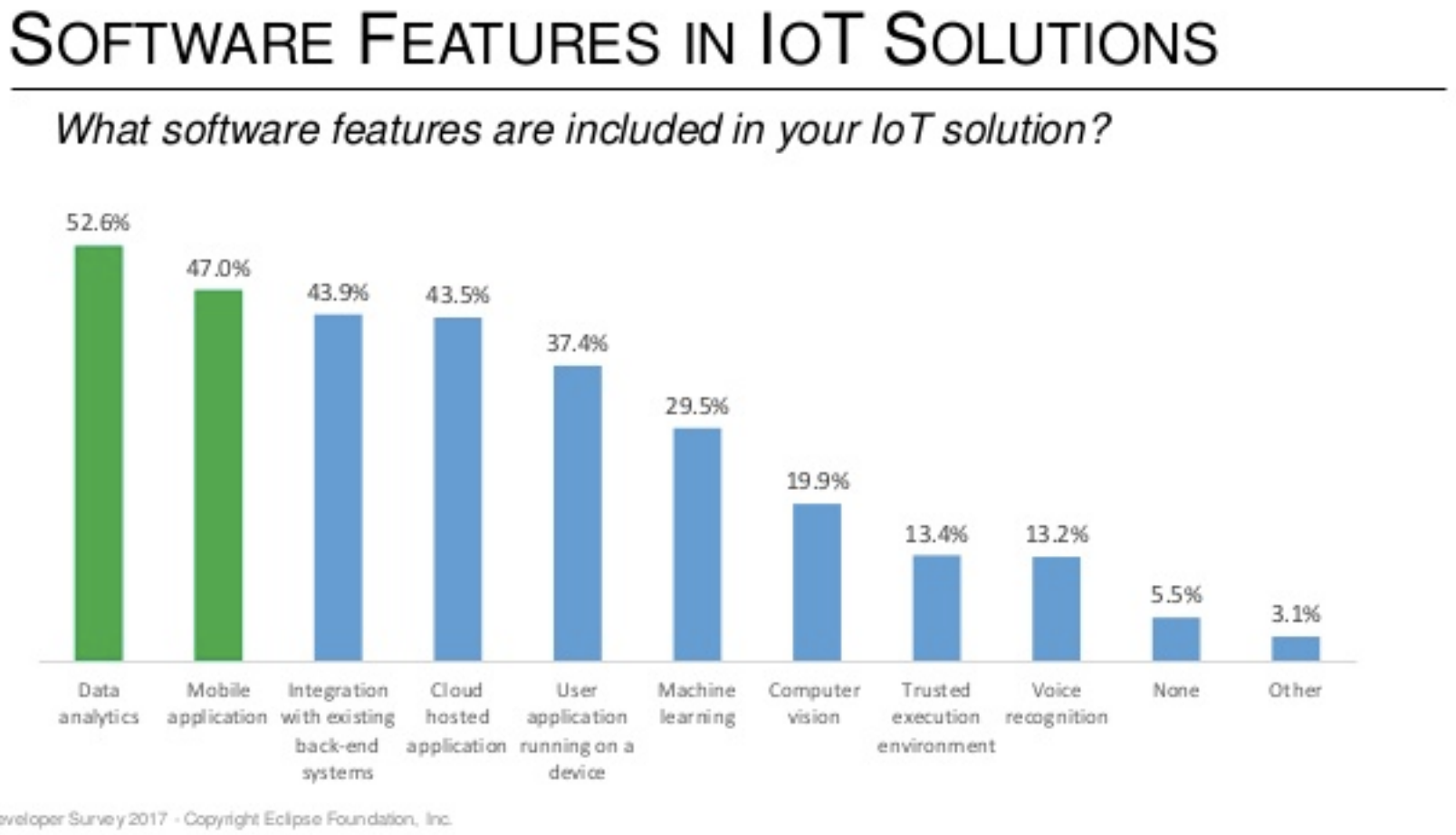
CLOUD SERVICES FOR IOT

Do you use, or plan to use, any of the following cloud service offerings for implementing your IoT solution?



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IoT Uygulamalarında Yazılım Özellikleri



KAYNAKLAR

- **Temel Kaynaklar**
  - **Ders Notları – Sunular**

- **Diğer Kaynaklar**
  - Eclipse Foundation IOT Developer Survey 2017